The role of white blood cells

There are five main types of white blood cells or *leukocytes* (*leuko* white/colourless, *cyte* cell). All leukocytes are formed from stem cells present in bone marrow. Depending on the cell type, the leukocyte will either mature while still in the bone marrow or, as in the case of the T-cells, they are formed in the bone marrow then travel to the thymus to mature. (The thymus is situated near the breastbone/tracheal region.)

There are fewer white blood cells in the blood than red blood cells. It is estimated that they are outnumbered by at least 700 to one. However, this does not mean they are found in small numbers! In fact, a conservative estimate would number hundreds of millions (probably more) though many die within 24 hours. Once formed, white blood cells begin circulating throughout blood, the lymph system and tissues.

When body tissue is damaged, *chemicals* are released that attract white blood cells to the site of the damage. Invading bacteria also cause the release of chemicals that attract white blood cells. The primary role of all leukocytes is to protect the body from invasion by disease-causing viruses and bacteria. White blood cells also perform general housekeeping duties by digesting dead and dying cells and other unwanted matter found within the body. The three main types of white blood cells are: *granulocytes, monocytes* and *lymphocytes.*

1 Give a definition for *stem cells.*

Up to 75% of the body's white blood cells are called *granulocytes* (*granum grain*) with estimates of 50–60% on average. Granulocytes get their name from the grainy substances found in the cytoplasm. Therefore, if it is a white blood cell with the grainy cytoplasm, regardless of the role it plays, it is classified as a granulocyte. There are several types of granulocytes and each is identified by the way it stains. These are:

- *Basophils,* which are identified by staining a pale colour. They are associated with antihistamine reactions, though this does not mean they perform no other role.
- *Neutrophils* work with the immune system's 'search and destroy' cells. They stain a neutral colour.
- *Eosinophils* play a major role in allergic reactions. They have also been identified working with other leukocytes, particularly with assisting chemical reactions. Eosinophils stain a reddish-brown colour and are believed to represent the smallest group of leukocytes with estimates at less than 5% of the total.

2 Give two reasons why scientists might stain cells.

As a whole, the different granulocytes play varying roles in the body's defence system and might therefore be considered as a group of 'all rounders'.

Monocytes are large cells known by two different names, depending on where they are found within the body at the time. They are called monocytes when circulating (and working) throughout the blood system. However, not all substances that must be dealt with are found in the bloodstream. At times, the monocyte is called to digest something (be it an invader or one of the body's own dead cells) that is found within tissues or other areas. When it leaves the bloodstream to pursue a target, it is then called a *macrophage* (*macro* large, *phage* to eat/digest). When it returns to the blood, it is again referred to as a monocyte. Some sources will state that the monocyte evolves into a macrophage but it is better to consider their duel roles as described above. Approximately 7–9% of leukocytes are believed to be monocytes.

3 Which of the white blood cell types is the most abundant?

Antibodies and antigens

Not all white blood cells fight disease by digesting invaders. The *lymphocytes* do their bit by converting globulins (proteins found in blood plasma)* into antibodies. Antibodies are molecules that bind to invading substances thus 'tagging' them for destruction by the macrophages. The targeted substances are known as *antigens.* An antigen is often recognised as a protein found on a bacteria or virus 'outer coat' and it causes the production of an antibody. Depending on the type, antibodies can destroy or inactivate bacteria and viruses or bind to the antigen and 'flag' it for the macrophage to deal with.

* There are a number of different globulins found within the blood plasma.

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Antibodies are very specific. Once a person has had a particular infection the antibodies produced will stay dormant in the body but will be reactivated whenever that particular antigen reappears. This gives a certain amount of immunity to subsequent infections from the same type of microbes. It also saves time because the immune system can act faster if it has already dealt with the antigen.

4 What is blood plasma?

5 How is the name *macrophage* derived?

6 Give a definition for a *globulin*.

7 Define *dormant*.

8 Explain the difference between a *monocyte* and a *macrophage*.

B-cells and T-cells

There are two main sub-groups of lymphocyte: B-cells and T-cells. The *B-cells* are responsible for antibody production. When an invader is recognised, antibodies are produced from proteins present in blood plasma. The antibodies then attach to the non-self invaders, flagging them for destruction. Like all lymphocytes, B-cells are formed in the bone marrow and this is where they mature. In other words, once they leave, they are ready to work.

There are three types of *T-cells*. As the name suggests, the *Killer T-cells* are the 'search and destroy cells'. These cells destroy invaders and dead or damaged 'self' sells that are recognised as unfit – including some forms of cancer cells. While other forms of treatment are usually required to fight cancers, Killer T-cells abound where it is present. T-cells form in the bone marrow but must migrate to the thymus to complete maturation. Once mature, many will head for the lymph tissue while others will circulate throughout the bloodstream.

9 The lymph glands are positioned at strategic parts of the body to enable a quick response by the glands closest to infection. Suggest how storing a number of T-cells in lymph tissue at all times is of benefit to fighting disease.

Two 'brothers' to the Killer T-cell are the *Helper T-cell* and the *Suppressor T-cell*. Helper T-cells activate the required immune system responses to warn the Killer T-cells and the B-cells that trouble is brewing (*'Wake up! We're being invaded!'*). The Helper T-cells cause the existing population of Killer T-cells and B-cells to rapidly and repeatedly divide, thus creating a cloned army that will set out to attack the invaders. Later, as the activity winds down and the 'army' is no longer needed, the Suppressor T-cells slow down the activity that the original immune response activated (*suppress* to restrain or slow down).

10 Suggest what might happen if the Suppressor T-cells did not slow down the Killer T-cells and the B-cells after the invading organisms had been dealt with.

11 What does the following describe: *'A molecule that initiates an antibody reaction.'*?

12 Summarise the roles of B-cells and T-cells.

13 Describe the difference between an *antibody* and an *antigen*.

14 **Mini project!** Draw a small, cartoon-style picture illustrating the roles of the different types of white blood cell.

Did you know? The gooey substance we call *pus* is made up of dead tissue and dead bacteria but mostly dead leukocytes – most of which are neutrophils.

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